Group Project: Credit Application Governance Analysis

Data Ecosystems and Governance in Organizations
Group 8

Msc Business Analytics
Nova SBE

Business and Project Context

THE PROBLEM

NovaCred received a **regulatory inquiry about potential discrimination** in their ML-driven credit decisions.

THE RISKS

- GDPR fines up to €20M or 4% of global turnover
- EU AI Act: credit scoring HIGH-RISK AI system (Annex III)
- Reputational damage and loss of operating licence

01 **Data Quality Audit**
Completeness, consistency, validity & accuracy

02 **Bias Detection**
Disparate impact, age bias, proxy & interaction

03 **Privacy & Governance**
PII mapping, pseudonymization, GDPR & AI Act

01 **Conclusion**
Key metrics & next steps

Team Presentation

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Dataset Overview and Objective

502
Raw Records

500
Clean Records

15
Columns Audited

3
Analyses Run

Objectives

- Audit credit applications for data quality
- Detect algorithmic bias & discrimination
- Propose GDPR & EU AI Act controls

Data Quality-Issues

Completeness

Issue

- Missing values were detected in key applicant attributes.
- Incomplete demographic data may affect fairness analysis and age-based bias detection.

Evidence

- notes: **100% missing**
- loan_purpose: **90% missing**
- processing_timestamp: **87.6% missing**

- **~20% cells missing overall**

Remediation

- Missing values were identified using **null checks**.
- Records with missing demographic attributes were **flagged.**
- These cases were handled during **preprocessing to prevent bias** in fairness analysis.

Data Quality-Issues

Consistency

Issue

- Inconsistent encoding was detected in categorical and date fields.
- Different formats can create artificial categories and lead to incorrect analytical results.

Evidence

- Gender field contained multiple encodings:
  - Male / Female
  - M / F
- A total of **111 records (22.1%) used abbreviated gender values.**

- Three different **date formats** were also detected:
  - YYYY-MM-DD **(67.9%)**
  - DD/MM/YYYY **(20.2%)**
  - YYYY/MM/DD **(11.2%)**
- **5 records as 'annual_salary'** insted of 'annual_income'

Remediation

- Gender values were standardized into two consistent categories:
  - **male** and **female**.
- Date fields were normalized using **pd.to_datetime()** to ensure consistent parsing across all records.

Data Quality-Issues

Validity

Issue

- Invalid financial values were detected that violate logical domain constraints for credit applications.

Evidence

- Examples of **invalid entries** included:
  - annual_income stored as **str, int & float**
  - savings_balance with **1 negative** records
  - credit_history_months with **2 negative** records
- These values are **not logically possible** in a financial application context.

Remediation

- Validation rules were applied to detect invalid values.
- Invalid values were replaced with **NaN.**
- Missing numeric values were imputed using the **median**, which is robust to outliers.

Data Quality-Issues

Accuracy

Issue

- Conflicts and duplication were detected in identity information, which may affect the reliability of the dataset.

Evidence

- **2 duplicate application IDs** were identified.
- **4 SSN conflicts** were detected where the same SSN appeared across different applicants.
- 4 invalid email addresses registered **(0.80%)**

- These inconsistencies may indicate data integration errors or potential identity fraud.

Remediation

- Duplicate records were **removed** from the dataset.
- SSN conflicts were **flagged for investigation**.
- Integrity checks were introduced to prevent duplicate identifiers in future data ingestion processes.

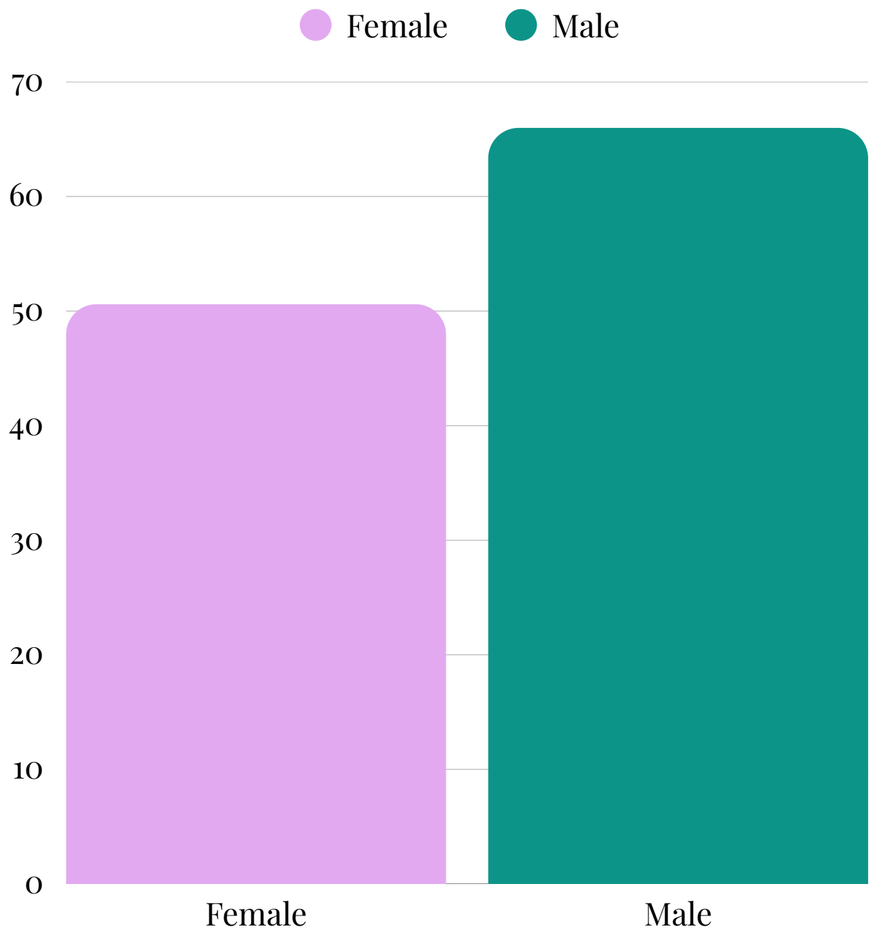
Bias Detection

DI Ratio

0.767

<0.8 ↠ BIAS DETECTED

Four-fifths rule violated



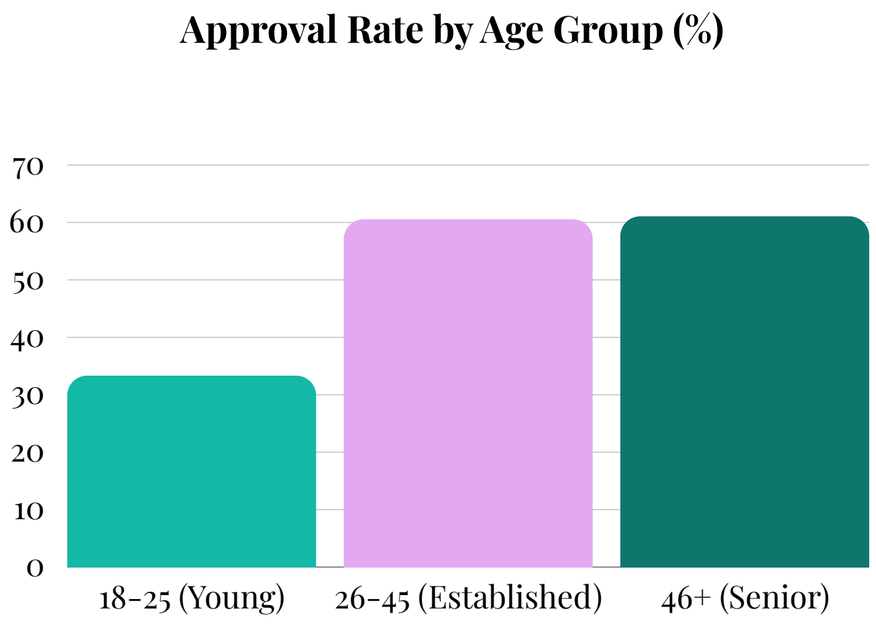
Key Insights

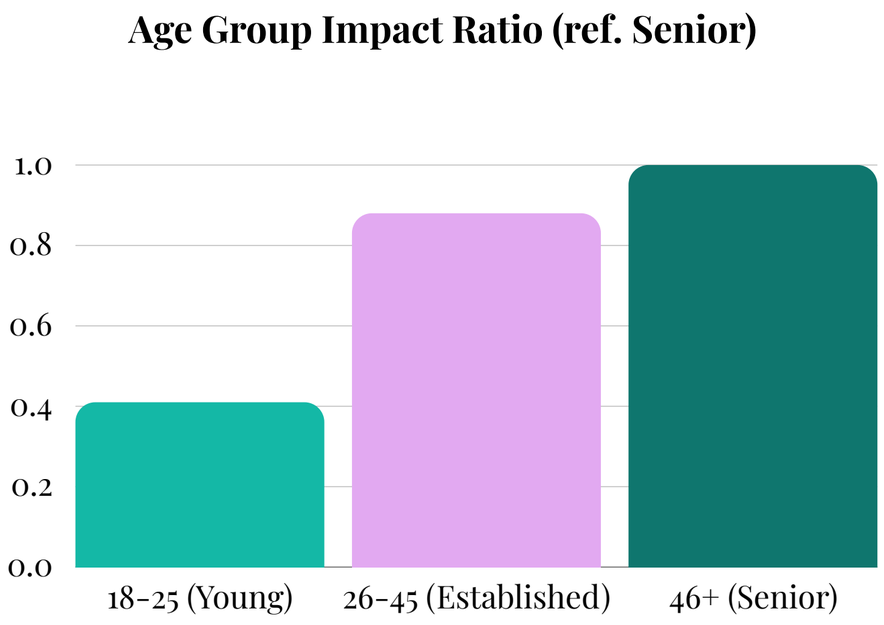
- Women: **50.6%** approval vs Men: **66.0%**
- 15.4 pp gap despite similar financial profiles
- Approved women have **8.5%** higher income than approved men
- Approved women need **10.2%** longer credit history
- Pattern suggests hidden selection bias

Finding

- Women require significantly **stonger financial profiles to obtain the same approval outcome** — consistent with EU AI Act Art. 10 requirements on data quality and non-discrimination.

Age Bias & Proxy Variable: Credit History

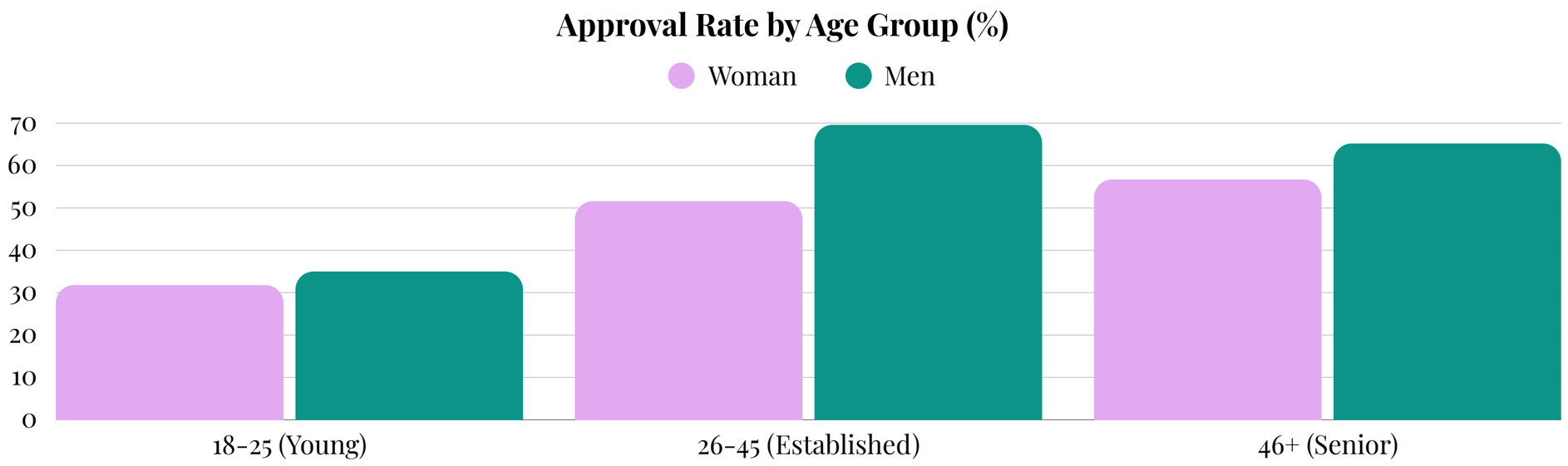




Proxy Discrimination:

- credit_history_months correlates with age at **r = 0.65**
- Younger applicants are structurally penalized regardless of their actual creditworthiness.

Intersectional Bias: Who is more disadvantaged?



31.8%
Young Women Approval Rate
Lowest of all 6 groups

-30.2pp
Gap vs Senior Men (65.2%)
Compounded Penalty

35%
Young Man Approval Rate
2nd lowest group

Key Insight - Interaction Effects

- Young women **(31.8%)** face compounded disadvantage — penalised for both age and gender simultaneously, producing the lowest approval rate of all six groups.
- The **30.2 pp** gap between Young Men **(35%)** and Established Men **(65.2%)** confirms age as an independent discriminator, further amplified for women — a clear intersectional pattern requiring governance intervention.

All PII Fields Identified & Classified

Field	Category	Risk	GDPR Basis	Control
full_name	**Direct Identifier**	**Critical** 🔴	Art. 5, 25	SHA-256 presudonymization
email	**Direct Identifier**	**Critical** 🔴	Art. 5, 25	SHA-256 presudonymization
ssn	**National Identifier**	**Critical** 🔴	Art. 9, 32	Salted hash + access control
ip_address	**Online Identifier**	**High** 🔴	Art. 4(1), 25	Pseudonymize; 30-day retention
date_of_birth	**Quasi-Identifier**	**Medium** 🟠	Art. 5(1)(c)	Generalize → age band
zip_code	**Quasi-ID / Proxy**	**Medium** 🟠	Art. 5(1)(c)	Truncate → ZIP3
gender	**Protected Attribute**	**High** 🔴	Art. 9,22	Audit-only; excluded from model
spending_behavior	**Behavioral Data**	**High** 🔴	Art. 5(1)(c)	Aggregate; remove sensitive categories

Privacy Protections Applied

Pseudonymization

SSN (salted SHA-256)

```
BEFORE: 596-64-4340
AFTER:
ssn_523f694b224accec
```

Email (salted SHA-256)

```
BEFORE: jerry.smith17@hotmail.com
AFTER:
email_83df9c1280f05445
```

Data minimization

```
DOB:  2001-03-09  →  age band: 18-24
ZIP:
10036        →  ZIP3:       100
```

Method: HMAC-SHA256 with environment salt — deterministic (linkable for audits) but non-reversible without the key.
Satisfies **Art. 25 (Privacy by Design) & Art. 32 (Security of Processing).**

Governance Gaps Identified

Unprotected PII
SSN, email, name stored as plain text → pseudonymize all direct identifiers

No Consent Tracking
No lawful basis documented → implement consent log (Art. 6 & 7)

No Retention Policy
Data stored indefinitely → enforce 5-year max + deletion workflow (Art. 5(1)(e))

No Audit Trail
Zero traceability of decisions → add immutable decision log (Art. 30)

No Human Oversight
Fully automated credit decisions → require human review flag (AI Act Art. 14)

Sensitive Behavioral Data
Gambling/Adult Ent. categories collected → aggregate or drop (Art. 5(1)(c))

GDPR Mapping, EU AI Act & Governance Controls

Field	Principle	Control Applied	Status
Art. 5(1)(b) & (c)	Purpose limitation & data minization	Drop notes; generalize DOB & ZIP; aggregate spending	**Implemented**
Art. 6 & 7	Lawful basis & consent	Consent tracking mechanism required	**Proposed**
Art.9	Special category data (gender)	Gender excluded from model; audit-only use	**Implemented**
Art.17	Right to erasure	Right-to-erasure deletion workflow required	**Proposed**
Art.22	Automated decision-making	Human review flag for credit denials	**Proposed**
Art.25	Privacy by design & by default	Pseudonymization & minimization by architecture	**Implemented**
Art.30	Records of processing activities	Audit trail of all governance actions	**Implemented**
Art.32	Security of processing	Salted SHA-256 hashing; environment-variable salt	**Implemented**
Art. 5(1)(e)	Storage limitation	5-year max retention + automated deletion policy	**Proposed**

EU AI Act — HIGH-RISK Classification (Annex III, §5b

- **Credit scoring: HIGH-RISK AI** → affects access to financial services
- **Mandates:** risk assessment, transparency, human oversight (Art. 14), data governance docs (Art. 10 & 11)
- Non-compliance before deployment = prohibited market entry

Key Takeaways

Data Quality	• 502 → 500 records • 8 issue categories resolved across completeness, consistency, validity & accuracy.
Bias Confirmed	• DI Ratio = **0.767 (<0.8)** violates the four-fifths rule. • Young adults face an impact ratio of 0.41. • Credit history acts as an age proxy (r = 0.65).
Privacy Governance	• 7 PII fields identified; pseudonymization implemented for SSN, email, name & IP. • 5 governance controls proposed.
Next Steps	• Retrain model with bias-corrected data, implement consent & retention policies, achieve EU AI Act compliance before deployment.